

1 Debunking decreases mistaken causal inferences: A direct replication of Irving et al. (2022)

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Abstract

We conducted a direct replication of Irving et al. (2022) examining whether a debunking statement reduces mistaken causal inferences from correlational evidence. We randomly allocated participants to read a news article containing either a correction statement ($n = 65$) or a no-correction control ($n = 64$). Participants in the correction group made fewer causal inferences ($\delta = 0.65$, 95% HDI [0.30, 1.00]) and more correlational inferences ($\delta = 0.72$, 95% HDI [0.36, 1.07]) compared to the no correction group. In exploratory analyses, we also supported the robustness of their findings to alternative modelling choices. Our findings represent a successful direct replication of Irving et al.

Keywords: statistical misinformation, causal inferences, debunking, replication, reproducibility

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26 **Social Impact & Responsibility**

27 Irving et al. (2022) investigated how debunking statistical misinformation can reduce
28 mistaken causal inferences. Empirical findings may be presented in public domains using
29 non-scientific language or inadequate statistical reporting which can lead to mistaken
30 inferences. A common risk with such misinformation is mistaking a correlational finding for
31 a causal one. Such mistaken inferences may surpass public opinion and potentially influence
32 domains such as policy decision making and medical research. This may be more salient for
33 populations where the original scientific source is inaccessible. Our success in replicating the
34 results of Irving et al. (2022) suggest that the debunking effect is robust and the inclusion of
35 a correction statement can reduce the risk of mistaken causal inferences. However, our study
36 only compared debunking to a control, so we cannot make any conclusions about the
37 effectiveness compared to alternative techniques.

Misinformation affects the whole spectrum of society from everyday conversations, to the media, to publishing scientific research. Misinformation is where someone makes a claim or shares a piece of information they initially believe to be true but later turns out to be false (Lewandowsky et al., 2012). In contrast, disinformation is where someone intentionally shares information they know to be false (Hameleers, 2023). Though unintentional, misinformation may lead to misinformed policy decisions and downstream changes in individual behavior and beliefs. For instance, Loomba et al. (2021) found that exposure to misinformation led to lower vaccination intent among residents across the USA and the UK during the COVID-19 global health crisis.

Debunking can be an effective strategy to counter misinformation where corrective messages are included to caution viewers (Chan et al., 2017). While inherent attitudes and beliefs may be resilient (Jang et al., 2019; Schmid, 2026), debunking can reduce mistaken inferences (Ecker et al., 2022; Fasce et al., 2026). In the current study, we focus on misinformation around media articles reporting on scientific research reports and whether that information can be debunked. We present a direct replication of Irving et al. (2022) who found that debunking statistical misinformation through a caveat statement has a large effect on reducing the number of mistaken causal inferences. Given the importance of reducing the impact of misinformation, it is important to independently replicate and verify the robustness of their results to ensure a trustworthy foundation for future research.

Debunking statistical misinformation in the media is particularly important as lay readers may not have the quantitative skills to debunk information on their own. Journalists are an easy target to blame for introducing misinformation, but Sumner et al. (2014) found 33% of university press releases exaggerated causal claims in research and 81% of articles using those press releases contained a similar exaggeration. In a follow-up, Adams et al. (2019) manipulated press releases to include more cautious headlines and explicit caveat statements around concluding causality, which led to headlines that were better aligned with

the evidence. Conversely, Irving et al. (2022) focused on the readers of news articles to see whether debunking statements affected what participants would understand about the study. Participants completed a continuous influence paradigm where they read a fictitious news story one sentence at a time. The story reported a new study showing extended TV screen time causes cognitive decline in older adults. Irving et al. manipulated one key sentence where participants were randomly allocated to read one of two versions. In the correction group, the story included a quote from the study's author warning they only demonstrated a correlational, not causal link. In the no correction group, the story stated they could not reach the author for comment. Irving et al. found strong evidence that correcting statistical misinformation decreased the number of mistaken causal inferences ($\delta = 0.64$ as a standardised mean difference) and increased the number of correlational inferences ($\delta = 1.27$) compared to the no correction group. Given the importance of developing impactful interventions to combat statistical misinformation, it is important to check the consistency of this finding.

Choice of Study for Replication

The ability to reproduce and replicate a study is critical to science, where reproducibility refers to taking the same data and applying the same analyses as an original study, and investigating whether someone can recreate the results (National Academies of Sciences, Engineering, and Medicine, 2019 (NASEM)). On the other hand, replicability involves collecting new data to address the same research question with the same research design as an original study (Nosek & Errington, 2017). While reproducibility is a minimum standard for scientific research, interpreting replicability is more nuanced. Successfully reproducing a study can increase confidence in the finding and establish a more robust basis for future research (Zwaan et al., 2018). But if a study does not replicate, there are helpful and non-helpful resources of replication failure (NASEM, 2019). Helpful sources include previously unknown contextual moderators while unhelpful sources range from mistakes to

fraud. Despite the importance of replication in science, they are still rarely published, representing an estimated 1% or less of articles (Clarke et al., 2026; Makel et al., 2012).

Selecting a study to replicate often comes from a place of suspicion where authors are skeptical about whether a study will replicate. Field et al. (2019) highlight three facets to select a study for replication when resources are scarce: statistical, theoretical, and methodological. These include whether the study's results are based on weak statistical evidence, whether the study addressed a theoretically important topic or currently relevant research question, and elements of the study's method like the sample size. Isager et al. (2023) proposed a formal decision model to describe this process and calculate replication value as a product of value and uncertainty of the original claim. Value relates to the scientific or societal impact of the findings and uncertainty relates to quantitative indicators like the sample size or statistical evidence. When researchers rated study attributes that would affect their confidence in whether a study would replicate, the attributes with the highest decrease in confidence included: whether the study had low statistical power, the methods and data were not available, and weak statistical evidence like a p -value close to .05 (Alister et al., 2021). For these considerations of replication value, there is a large component dedicated to the negative features of a study to motivate a direct replication.

In contrast, we selected Irving et al. as our replication target due to its positive features. Through the frame of Field et al. (2019), theoretically we consider correcting statistical misinformation to be a contemporary research question worth addressing; statistically, Irving et al. found very strong evidence for their effect; and methodologically, they justified their sample size through a power analysis, pre-registered their study, and shared their materials and data. By replication value (Isager et al., 2023) we consider Irving et al. to have high value given the societal importance of reducing the impact of misinformation, while having low uncertainty due to their robust findings. Alister et al. (2021) found that high statistical power, open materials, and study pre-registration were

among the attributes that most increased researchers' confidence in replication. In sum, we identified Irving et al. as an article to check the reproducibility and replicability of their finding that correcting misinformation reduced mistaken causal inferences.

Research Question and Hypotheses

Our research question is: Do the findings of Irving et al. (2022) replicate in a study independent from the original research team? There were two primary hypotheses which are the focus of our direct replication:

H1. Participants in the correction group would make fewer causal inferences than those in the no-correction group, in response to the coded inference questions.

H2. Participants in the correction group would make more correlational inferences than those in the no-correction group, in response to the coded inference questions.

The original study found strong evidence in favour of both primary hypotheses, and for the purposes of our direct replication, we pose the same hypotheses.

Method

Transparency and Openness

All materials, data, and analysis scripts are publicly available on our Open Science Framework (OSF) project: <https://osf.io/jpxbu/>. We included a .qsif file to import our study materials into Qualtrics and a PDF version. We used the statistics software R (R Core Team, R version 4.6.0 (2026-04-24)) for all analyses. This manuscript is available as a reproducible R Markdown document, and we provide a full report outlining our reproducibility checks of Irving et al. and outlining all our data processing and modelling steps. This study was not preregistered as we collected the data as part of an MSc dissertation project and we missed the opportunity due to the short turnaround of projects. We report how we determined our sample size, all data exclusions (if any), all manipulations, and all measures in the study.

Sample Size Justification

In the original study by Irving et al., they reported an *a priori* power analysis to detect a smallest effect size of interest of $d = 0.54$ based on the meta-analysis by Chan et al. (2017). They aimed to collect 75 participants per group ($N = 150$, $\alpha = 5\%$, $\text{power} = 95\%$), but their final sample size was 129, making their study sensitive to effects of $d = 0.58$ with 95% power. The smallest effect size reported by Irving et al. was $\delta = 0.64$ as roughly equivalent to Cohen's d , and if we use that in an *a priori* power analysis, we would need 54 participants per group ($N = 108$) for 95% power. As a replication study, we did not want to aim for fewer participants than the original study, so we recruited the same sample size of 129 to be sensitive to effects of $d = 0.58$ with 95% power.

Participants

With our target sample size of 129, we recruited participants online through Prolific between the 8th and 9th of June 2023. We paid participants £2.25 (prorated £9 per hour) for an estimated study time of 15 minutes. Our only sampling restriction was participants should be in the United Kingdom or the United States of America.

Given the online recruitment, we included a data quality question “We want to ensure that all data we analyse is meaningful and accurately reflects engagement with the study content. In your honest opinion, should we use your data in our analyses of this study?”. We removed one participant who answered “No, do not use my data” and recruited one additional participant to ensure the final sample size was 129 and consistent with our sampling plan. There were 65 participants in the correction group and 64 in the no correction group.

Table 1 shows the demographic breakdown of our sample compared to the original study by Irving et al. Our study was approximately five years younger on average and had more male participants, but the education profile and English as a first language was similar.

Table 1

Participant demographic characteristics in the current study compared to the original by Irving et al.

Demographic Characteristic	Aggarwal et al.	Irving et al.
Gender	Male = 72, Female = 56, other = 1	Male = 47, Female = 82
Age	39.83 (12.31)	44.76 (16.70)
Education Level	Bachelor's = 60 (46.51%), MSc = 22 (17.05%), PhD = 5 (3.88%), A-Levels or less = 42 (32.56%)	Bachelor's degree = 49.6%, MSc = 24.8%, PhD = 4.7%, A-levels or less = 20.9%
English as first language	English = 122, Not English = 6, NA = 1	English = 127, Not English = 2

Materials and Procedure

We created the study using Qualtrics based on the original study's open materials. The open materials allowed us to replicate the study as close as possible, where the only features not documented by Irving et al. were page breaks; and we used our own institution information sheet, consent form, and debrief. We had no contact with any of the original authors during the design of our study. The study received ethical approval from the University of Glasgow Medical, Veterinary, and Life Sciences ethics review committee (Ref: 200220281).

After being redirected from Prolific, participants first read an information sheet explaining the purpose of the study. Participants then provided informed consent and completed demographic questions on age, country of origin, gender, highest education level, and whether English was their first language. Participants completed a 21-item Trust in

Science and Scientists scale (Nadelson et al., 2014). Participants responded on a 1-5 Likert scale ranging from strongly disagree to strongly agree. After reverse coding 12 items, we calculated the mean of the 21 items to create an overall scale score, where higher values indicate higher trust in science and scientists. We included a 22-item spirituality scale (Delaney, 2005) to create doubt in the participant's perception of the study's aim. Like Irving et al., we did not process or analyse the participants' responses to this scale.

The critical materials for the study featured a continuous influence paradigm where participants read 10 short passages. Participants read each passage individually and could advance - but not go back - by clicking an arrow to show the next passage. The passages described a fictitious news article of a new study reporting to show the causal link between the number of hours someone watches TV and increased cognitive decline in adults over 65. The passages include fictitious quotes from a health and style expert, a member of parliament, and spokesperson from an online streaming service.

The independent variable controls the ninth passage where participants were randomly allocated to read one of two variations. Participants in the correction group read a passage which corrects the mistaken causal inference: "The lead author of the study, Professor Alastair McKay from Durham University, has since specified that the results have been misrepresented by Professor James Evans and that TV screen time has not yet been found to cause cognitive decline. He informs us that only a correlation has been found between the two and no causal link is stated in the study". In contrast, participants in the no correction group read a passage which does not correct the mistaken causal inference: "The lead author of the study, Professor Alastair McKay from Durham University could not be reached for comment".

After reading the 10 passages, we provided participants with 10 free-text responses asking them what they understood about the study. We manually coded 5 of the 10 passages for mention of a causal or correlational relationship (we provide further details of the coding

process below). Participants then completed 10 multiple-choice questions on facts about the passages, such as how old the participants in the news story were.

Participants completed a manipulation check question on whether they remember the results of the study being corrected or not and a statistical understanding check for whether they can explain the difference between correlation and causation. Unlike the original study, we included a self-report response check for whether we should have included their data in the analysis and the opportunity to leave a free-text response on their experience completing the study. Finally, participants read a debrief sheet fully explaining the aims of the study and the key manipulation.

Coding

As participants provided free-text responses, we had to manually code the responses for whether participants made a causal inference or a correlational inference. Author 3 (JB) prepared the responses by isolating the participant ID and inference questions from all other variables. This meant there was a level of blinding as the condition was not present while coding the data. Author 2 (KZ) then independently coded all responses by following the coding scheme Irving et al. shared with their pre-registration:

1. If a participant makes a correlational inference, they will receive a score of 0 on DV1 (causal inference) and a score of 1 on DV2 (correlational inference).
2. If a participant makes a causal inference, they will receive a score of 1 on DV1 (causal inference) and score 0 on DV2 (correlational inference).
3. If a participant gives an off-topic answer, they will receive a score of 0 on both DVs.

Like Irving et al., we coded responses using a harsh and lenient coding scheme. Under the harsh scheme, any ambiguous responses which did not directly comment on a causal or correlational relationship received 0. Under the lenient scheme, ambiguous responses for

questions three and four received 0 for causal and 1 for correlational if the participant answered “No” or “None”.

After Author 2 coded all responses, Author 1 randomly sampled the coding from 10 participants (50 individual codes). Any disagreements were resolved through discussion to provide a final spreadsheet of codes. For each participant, this meant there were four columns of data: five codes for causal inferences under the harsh and lenient scheme, and five codes for correlational inferences under the harsh and lenient scheme. Unless noted, our analyses focus on the harsh coding scheme to be conservative.

Replication Closeness Evaluation

Table 2 outlines replication classification criteria (LeBel et al., 2018) and we consider all design facets to be the same as the original study apart from two. We recruited people using Prolific whereas the original was a convenience and snowball sample using social media. We also paid participants whereas the original did not. Given the similarity in demographic variables (Table 1), we consider our study a very close direct replication under the replication taxonomy (LeBel et al., 2018).

Design and Data Analysis

Our study was a direct replication of Irving et al., meaning we collected the same variables and applied the same modelling techniques. To ensure we performed the analyses the same as the original study, we used the open data from Irving et al. to check computational reproducibility. We could reproduce their primary outcomes precisely, meaning we were confident we were using the same modelling techniques to ensure a direct replication.

In the online reproducibility and replication report, we provide all primary and secondary analyses that Irving et al. reported. For brevity, we focus on the primary

Table 2

Replication classification criteria (LeBel et al., 2018) comparing design facets of our study compared to the original by Irving et al.

Design Facet	Similarity	Reason
Effect / Hypothesis	Same	
IV Construct	Same	
DV Construct	Same	
IV Operationalisation	Same	
DV Operationalisation	Same	
Population	Different	We recruited participants online using Prolific. The original study recruited participants online through convenience and snowball sampling by sharing on social media.
IV Stimuli	Same	
DV Stimuli	Same	
Procedural Details	Same	
Physical Settings	Same	
Contextual Variables	Different	We paid participants through Prolific. The original instead study did not pay participants.

hypotheses in this manuscript, in addition to our own exploratory analyses to check the robustness of the findings to alternative modelling techniques.

The primary analyses focus on one independent variable (Condition) where participants were randomly allocated to the correction group or no correction group. There were two dependent variables for two ways of coding participants' inferences: 1) sum of causal inferences and 2) sum of correlational inferences. We calculated the sum of causal/correlational inferences by adding up the coded responses, where the values could range from 0 to 5.

To address the primary hypotheses 1 and 2, we applied a Bayesian one-sided independent samples t-test with a medium ($r = 0.707$) half-Cauchy prior using the BayesFactor R package (Morey & Rouder, 2023). For hypothesis 1, the one-sided test expected only positive effects for the no correction - correction comparison, meaning we expected participants in the correction group to produce fewer causal inferences compared to participants in the no correction group. For hypothesis 2, the one-sided test expected only negative effects for the no correction - correction comparison, meaning we expected participants in the correction group to produce more correlational inferences compared to participants in the no correction group.

To quantify the effect size of these analyses and compare to those included by Irving et al., we report the mean difference, delta (δ) as the standardised mean difference (Rouder et al., 2009), and the probability of superiority as a common language effect size. For the mean difference and delta, we report the 95% Highest Density Interval (HDI) as the Bayesian version of an interval estimate provided by the BayesFactor R package (Morey & Rouder, 2023). For the probability of superiority, we report the 95% confidence interval (CI) as the frequentist version of an interval estimate provided by the effectsize R package (Ben-Shachar et al., 2020).

Results

Direct Replication

Causal Inferences. On average, those in the no correction group ($M = NA$, $SD = NA$) made more mistaken causal inferences than those in the correction group ($M = NA$, $SD = NA$). Figure 1 shows the mean sum of causal inferences in the correction group compared to the no-correction group in our replication study compared to the original findings by Irving et al.

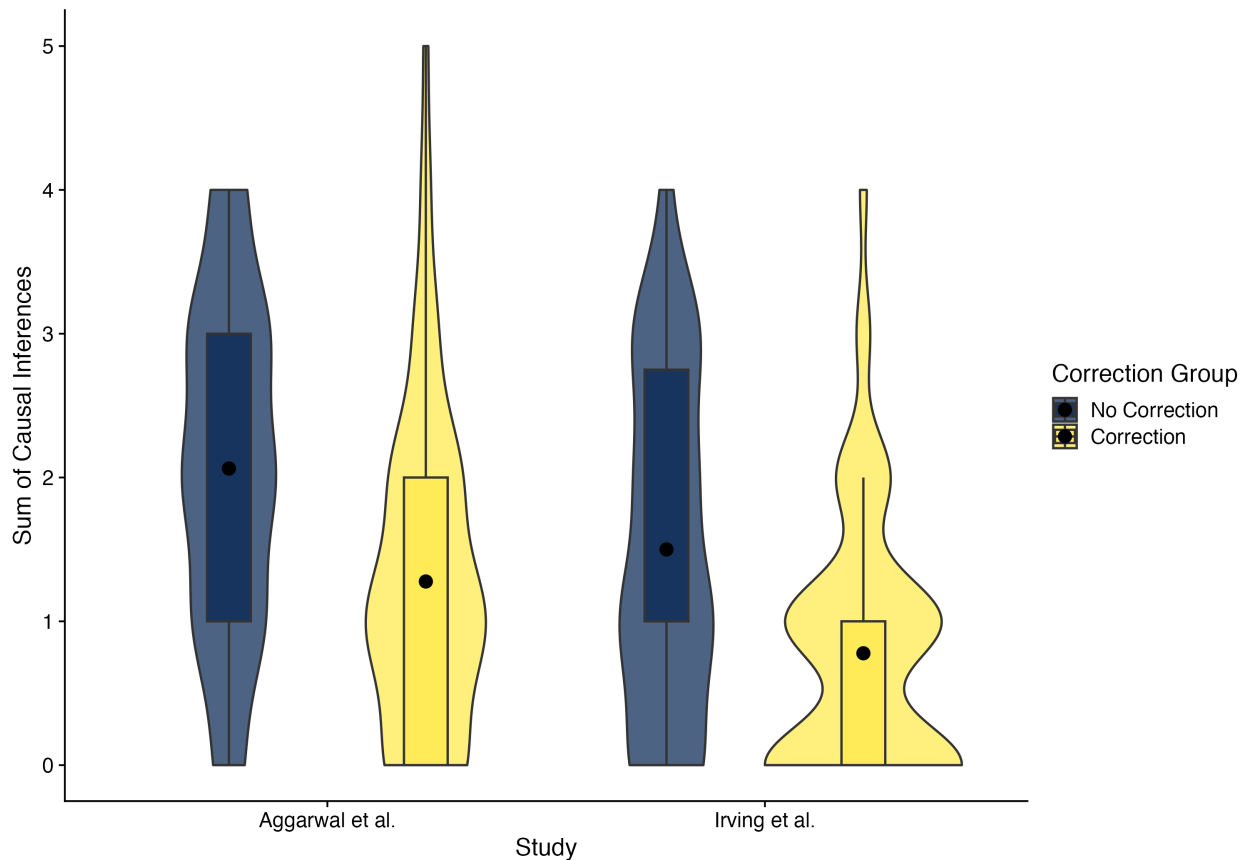


Figure 1. A violin-boxplot to compare the sum of causal inferences between the correction and no correction groups. On the left we visualise the findings from our replication study and on the right, we visualise the original data from Irving et al. for comparison. The violin element shows the density of responses, the boxplot element shows the distribution properties, and we add a circle for the mean rather than the standard median.

The Bayesian one-sided t-test showed the directional hypothesis was 308.69 times more likely than the null hypothesis. On average, participants in the correction group provided 0.74, 95% HDI = [0.35, 1.13] fewer causal inferences than participants in the no correction group, supporting H1. Expressed as a standardised effect size, this translates to a delta of 0.65, 95% HDI = [0.30, 1.00] and probability of superiority of 0.69, 95% CI = [0.59, 0.77]. This means there is a 69% chance that someone selected from random from the correction group would make fewer mistaken causal inferences than someone selected from random from the no correction group.

Correlational Inferences. On average, those in the correction group ($M = \text{NA}$, $SD = \text{NA}$) made more correlational inferences than those in the no correction group ($M = \text{NA}$, $SD = \text{NA}$). Figure 2 shows the mean sum of correlational inferences in the correction group compared to the no-correction group in our replication study compared to the original findings by Irving et al.

The Bayesian one-sided t-test showed the directional hypothesis was 1,212.41 times more likely than the null. On average, participants in the correction group provided 0.56, 95% HDI = [0.29, 0.84] more correlational inferences than participants in the no correction group, supporting H2. Expressed as a standardised effect size, this translates to a delta of 0.72, 95% HDI = [0.36, 1.07] and probability of superiority of 0.70, 95% CI = [0.61, 0.79]. This means there is a 70% chance that someone selected from random from the correction group would make more correlational inferences than someone selected from random from the no correction group.

Replication Evaluation. We used the statistical interpretation criteria from LeBel et al. (2019) to judge to what extent we replicated the findings from Irving et al. There are three criteria: (a) whether we detected a signal, (b) the consistency of the effect size, and (c) the precision of the effect size estimate compared to the original study.

For the sum of causal inferences, we consider the replication outcome to be signal

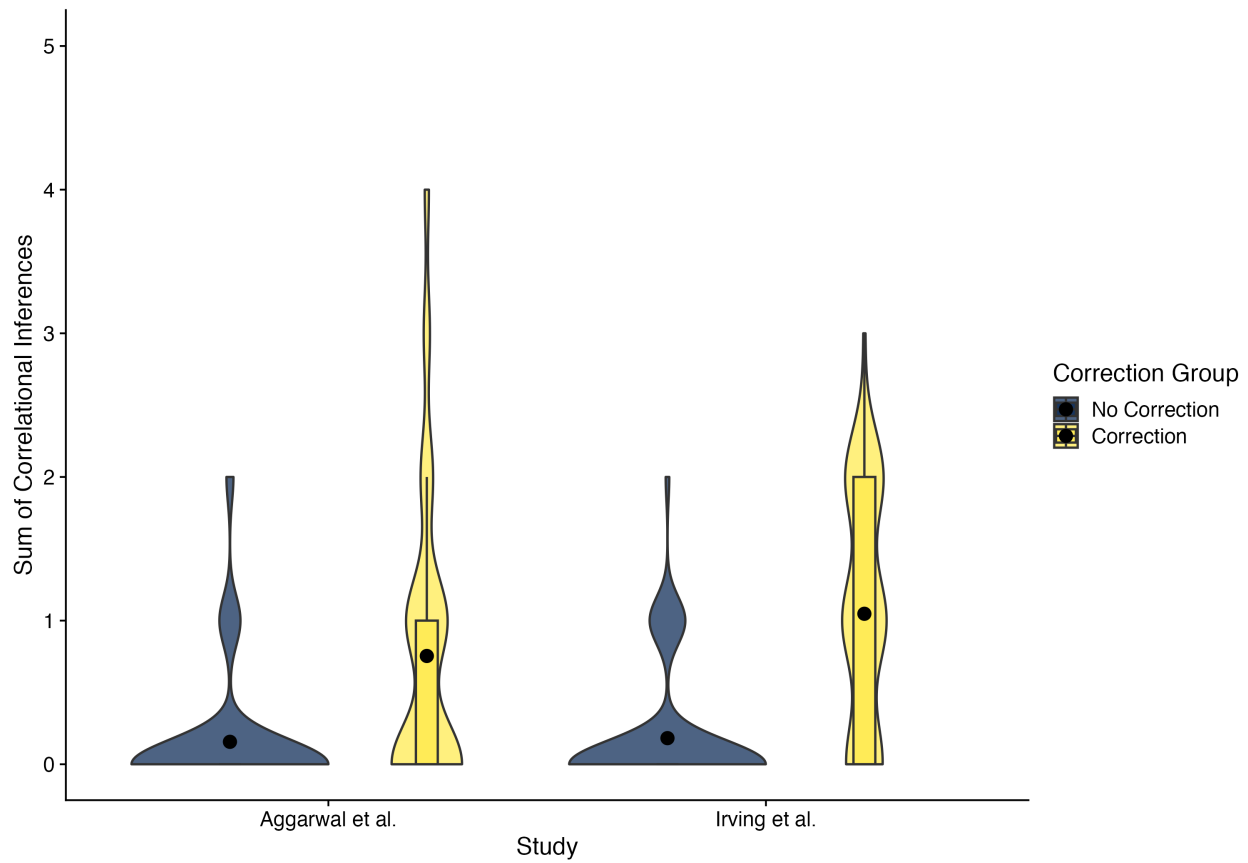


Figure 2. A violin-boxplot to compare the sum of correlational inferences between the correction and no correction groups. On the left we visualise the findings from our replication study and on the right, we visualise the original data from Irving et al. for comparison. The violin element shows the density of responses, the boxplot element shows the distribution properties, and we add a circle for the mean rather than the standard median.

detected, consistent with the original study. For the purposes of comparison, we focus on delta as a standardised effect size. Both our study ($BF_{10} = 308.69$) and Irving et al. ($BF_{10} = 279.35$) detected a signal (a), where the Bayesian one-sided t-tests found strong evidence in favour of the alternative hypothesis compared to the null. The estimate for delta (b) was almost identical between both studies ($\delta = 0.65$, 95% HDI = [0.30, 1.00] compared to $\delta = 0.64$, 95% HDI = [0.28, 0.99] in Irving et al. page 5). The 95% HDIs (c) were also similar between both studies, where the original effect size was within the replication 95% HDI, and the replication effect size was within the original 95% HDI. Figure 3 demonstrates this visually by comparing delta and its 95% HDI side by side, where the original and replication estimates overlap almost entirely.

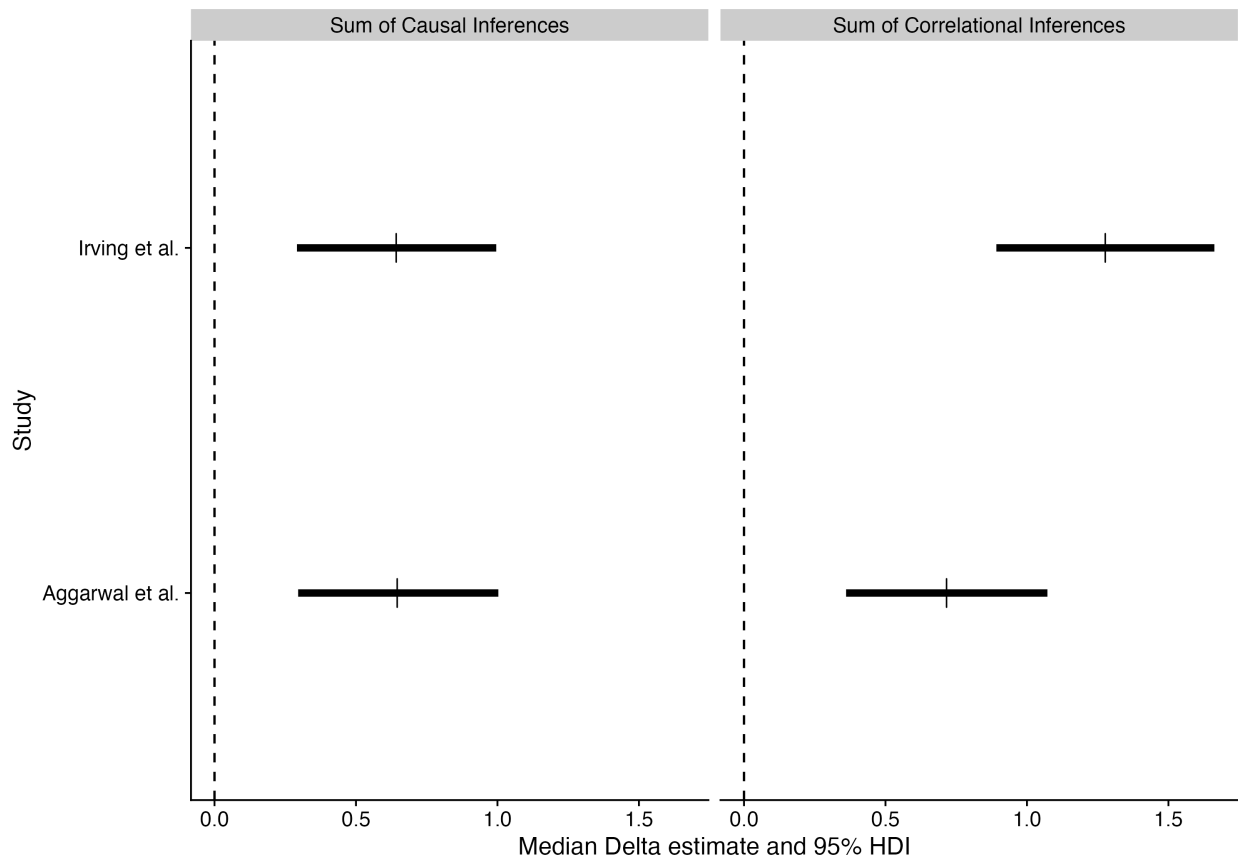


Figure 3. Comparing delta and its 95% HDI in the original and replication data for the sum of causal and correlational inferences.

For the sum of correlational inferences, the replication outcome is signal detected, inconsistent with the original study with a smaller effect size. Both our study ($BF_{10} = 1,212.41$) and Irving et al. ($BF_{10} = 1.36e+09$) detected a signal (a), where the Bayesian one-sided t-tests found strong evidence in favour of the alternative hypothesis compared to the null. The estimate for delta (b) was smaller in our replication ($\delta = 0.72$, 95% HDI = $[0.36, 1.07]$) compared to Irving et al. page 5 ($\delta = 1.27$, 95% HDI = $[0.90, 1.66]$), but in the same direction where those in the correction group reported more correlational inferences than those in the no correction group. The 95% HDI (c) in both studies had a similar range but the estimate in our replication shifted to a smaller magnitude closer to the sum of causal inferences estimates (see Figure 3).

Exploratory Analyses: Logistic Regression

Given the outcome distributions in Figures 1 and 2 that Irving et al. (2022) did not visualise, independent samples t-tests may not be the appropriate choice of analysis. The sum of causal and correlational inferences are on a bounded 0-5 scale, whereas the t-test assumes the outcome is normally distributed and theoretically boundless. Therefore, the analysis may not capture important features of the data generating process, particularly for the sum of correlational inferences where responses were close to floor levels. We checked the robustness of the conclusions by fitting a mixed effects logistic regression model using brms (Bürkner, 2017). The online reproducibility and replication report provides full details of the modelling approach and setting informed priors consistent with the WAMBS checklist for Bayesian modelling (Van de Schoot et al., 2021), but we provide an overview below of our final models.

Instead of the sum of causal/correlation influences, we modelled the outcome as binary using a Bernoulli distribution and logit link. For each inference question, the outcome was 1 if the participant made a causal/correlational inference and 0 if the participant did not make a causal/correlational inference. We included a categorical fixed effect for Condition to

compare the correction and no correction groups. We then included random slopes and intercepts for participant and question ID to take variation within participants and questions into account.

Causal Inferences. We used the open data from Irving et al. to fit our planned model with default priors and used the model estimates to set informed priors for our replication data. Note the following values are expressed on a log scale due to the logit link. For the intercept and slope priors, we used a normal distribution with a mean of -1.00 ($SD = 0.65$) for the intercept and a mean of -1.20 ($SD = 0.40$) for Condition slope. For the residual standard deviation prior, we used an exponential distribution with a rate parameter of 1. We fitted the model using four chains, each containing 4000 post-warmup samples (12,000 total) which passed model fitting and convergence checks.

The no correction group was set as the reference group and the mean estimate for the intercept was -0.51, 95% HDI = [-1.51, 0.50]. The correction group was set as the target group where the slope for Condition as a fixed effect was -1.16, 95% HDI = [-1.77, -0.55], suggesting lower log odds of making a mistaken causal inference in the correction group. To aid interpretation, we can convert the slope estimate from log odds to an odds ratio, where being in the correction group decreased the odds of making a mistaken causal inference 3.2, 95% HDI = [1.74, 5.85] times compared to the no correction group. Expressed another way, the predicted probability of making a mistaken causal inference was lower in the correction group (.16, 95% HDI = [.06, .35]) compared to the no correction group (.38, 95% HDI = [.18, .62]). Figure 4 compares the model predicted probabilities in our replication data using informed priors compared to the same model applied to the data from Irving et al. using default priors.

This shows the results were robust to the modelling approach as the correction group had lower odds of making a mistaken causal inference compared to the correction group, consistent with the sum of causal inferences being lower in the correction group compared to

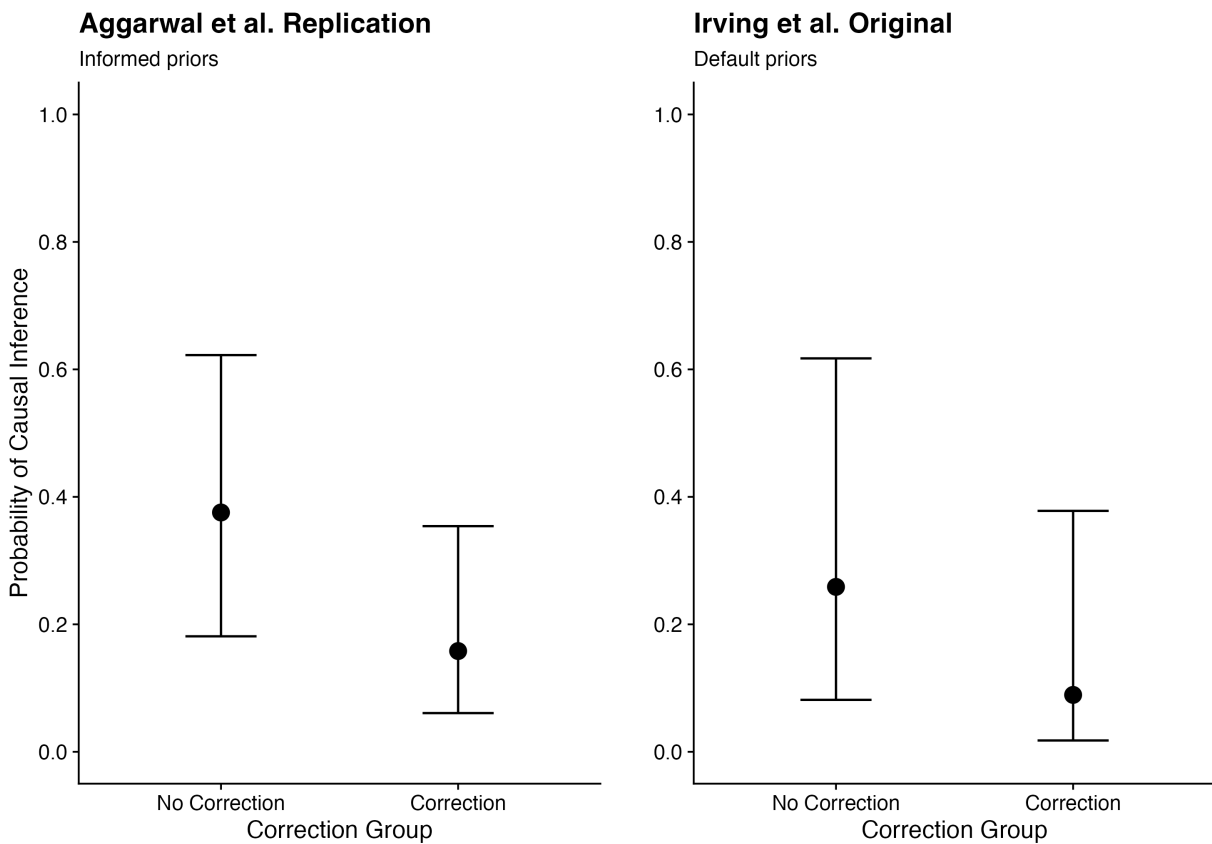


Figure 4. Lower predicted probability of making a mistaken causal inference in the correction group compared to the no correction group. Circles represent the mean estimate from the model, and the error bars are the 95% HDI around the estimates. The left panel shows the estimates from our replication study using informed priors and the right panel shows estimates from the same model applied to the original study by Irving et al. using default priors.

the no correction group.

Correlational Inferences. We fitted the same model to correlational inferences and the original data from Irving et al. to set informed priors. For the intercept and slope priors, we used a normal distribution with a mean of -3.40 ($SD = 0.5$) for the intercept and a mean of 1.9 ($SD = 0.6$) for Condition slope. With no correction as the reference group, we expect higher log odds in the correction group as the target. For the residual standard deviation prior, we used an exponential distribution with a rate parameter of 1. The model with informed priors passed model fitting and convergence checks.

The mean estimate for the intercept was -4.46, 95% HDI = [-5.42, -3.55] and the slope for Condition as a fixed effect was 1.90, 95% HDI = [1.00, 2.80], suggesting higher log odds of making a correlational inference in the correction group compared to the no correction group. Expressed as an odds ratio, being in the correction group increased the odds of making a correlational inference 6.71, 95% HDI = [2.72, 16.51] times compared to the no correction group. The probability of making a correlational inference was higher in the correction group (.07, 95% HDI = [.03, .16]) compared to the no correction group (.01, 95% HDI = [0, .03]). Figure 5 compares the model predicted probabilities in our replication data using informed priors compared to the same model applied to the data from Irving et al. using default priors.

This shows the results were robust to the modelling approach as the correction group had higher odds of making a correlational inference compared to the correction group, but we should be cautious in the interpretation. On the one hand, we can see clear advantages from using a logistic regression model compared to the analyses from Irving et al. There are few correlational inferences in the data with many zeros, so the estimates and lower bound of the HDI are close to zero predicted probability and respecting the bounded nature of the outcome. However, there is a lot of uncertainty in these estimates shown by the wide 95% HDI around the odds ratio and predicted probability estimates, potentially due to few correlational inferences in the data.

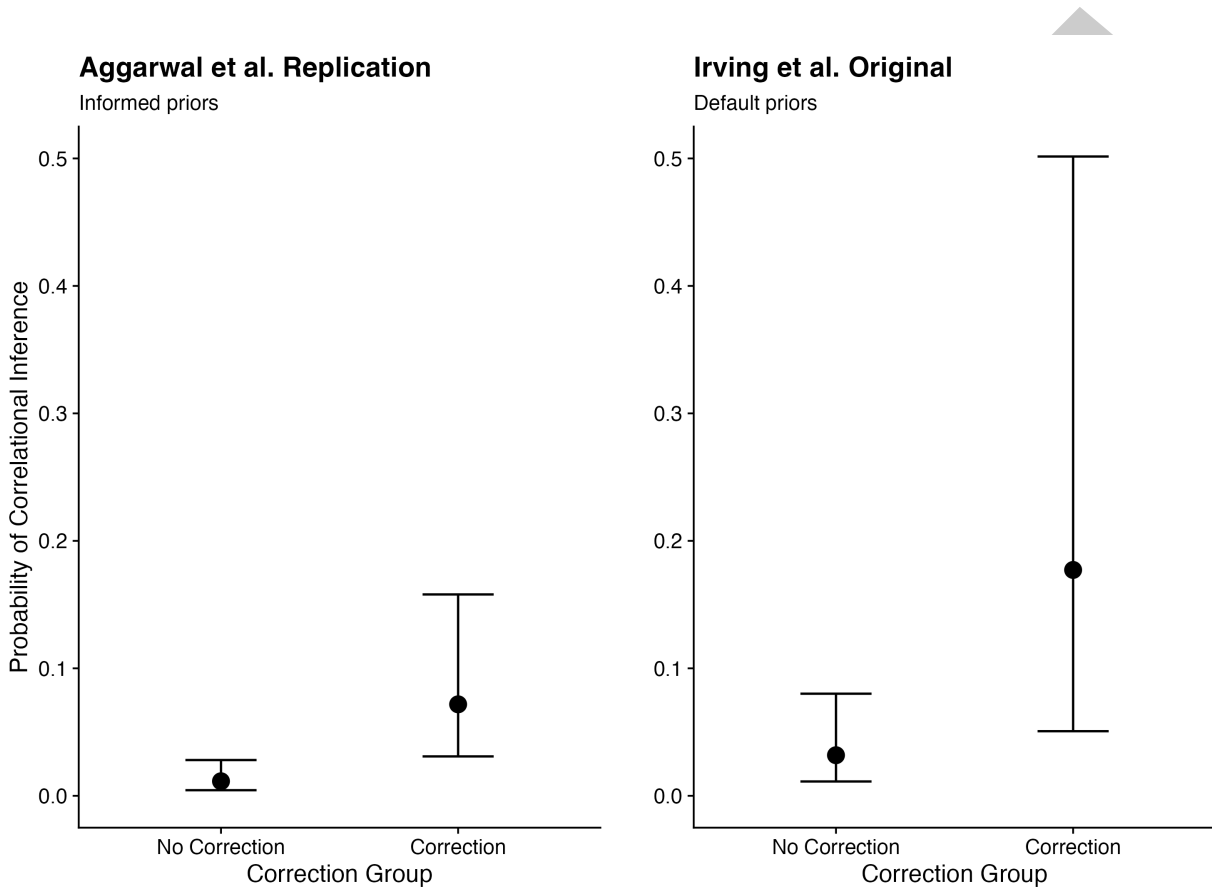


Figure 5. Higher predicted probability of making a correlational inference in the correction group compared to the no correction group. Circles represent the mean estimate from the model, and the error bars are the 95% HDI around the estimates. The left panel shows the estimates from our replication study using informed priors and the right panel shows estimates from the same model applied to the original study by Irving et al. using default priors.

Discussion

In this direct replication of Irving et al. (2022), we found debunking statistical misinformation lead to fewer causal inferences among readers of a news article. Using the replication success criteria of LeBel et al. (2018), we categorise causal inferences to be signal detected, consistent with the original study. The standardized effect size we observed ($\delta = 0.65$, 95% HDI = [0.30, 1.00]) was almost identical to the estimate reported by Irving et al. ($\delta = 0.64$, 95% HDI = [0.29, 0.99]). For the second outcome, we similarly found that debunking statistical information led to more correlational inferences, but we categorized the replication success criteria as signal detected, inconsistent with the original study with a smaller effect size. The standardized effect size we observed ($\delta = 0.72$, 95% HDI = [0.36, 1.07]) was smaller than Irving et al. reported ($\delta = 1.28$, 95% HDI = [0.89, 1.66]). The inconsistency is potentially due to the difference in manual coding for correlational inferences compared to the original research team, but the effect size being at least half the size of the target study is comparable to Lopez et al. (2023) in their replication of visual cues and food. We describe our findings as a successful direct replication where we confirmed both hypotheses H1 and H2, with the effect size for causal inferences being virtually identical and the effect size for correlational inferences being smaller.

The open research practices of Irving et al. (2022) allowed us to assess all four dimensions of reproducible research (The Turing Way Community, 2025): reproducible, replicable, robust, and generalisable. First, the study was computationally reproducible as we obtained identical results for the primary analyses by using open data from Irving et al. (2022). Second, the study was replicable as we were able to produce similar results in the primary analyses using their analysis approach on our newly collected data. Third, the original findings were robust when we applied an alternative analysis technique and made qualitatively similar conclusions. Finally, the findings were generalisable as we also made qualitatively similar conclusions using the alternative analysis technique on our newly

collected data. In sum, we consider the findings of Irving et al. to meet all the criteria for reproducible research.

Through this replication, we present evidence supporting the robustness and effectiveness of caveat statements as a debunking strategy for statistical misinformation. Misinformation is a growing phenomenon in today's digital world where information is consumed rapidly and continuously through mediums such as social media (Xu et al., 2025). This has further been exacerbated by the sharp increase in generative artificial intelligence content (Park & Nan, 2026). Our successful replication of Irving et al. (2022) supports the effectiveness of a debunking caveat statement in reducing misinformation across different topics (Lorko et al., 2026), consistent with the strategy of providing an alternative narrative in reducing the impact of misinformation (Lewandowsky et al., 2012). Debunking strategies such as caveat statements represent an effective tool to reduce misinformation, perhaps in conjunction with other strategies such as combining media literacy interventions and fact-checkers (Hameleers, 2023).

Limitations and Future Research

While we consider our findings to represent a successful direct replication of our target study, there are limitations to consider around external validity (Clarke et al., 2024). In terms of sample diversity, our replication was similar to Irving et al. (2022) where participants were predominantly educated to an undergraduate degree level or higher, and located in the United Kingdom or United States of America where English was their first language. Future research could explore the generalisability of caveat statements as a debunking strategy across more diverse samples to identify potential moderators, such as through a big-team science approach. Klein et al. (2018) explored the consistency of their multi-site replication against different dimensions of the WEIRD (Western, Educated, Industrialized, Rich, and Democratic) acronym but large data collection networks can still miss several dimensions of generalisability if participants are well-educated and restricted to

urban environments (Ghai et al., 2024). Therefore, while we replicated the original findings with a similar sample, future studies could explore the generalisability of caveat statements for debunking statistican misinformation across cultures or socio-demographically diverse samples.

A second limitation relating to external validity is the low realism of the task and online data collection. Participants completed a continuous influence paradigm where they read a fictitious news article one sentence or passage at a time. We included a free-text box prior to the debrief to offer participants an opportunity to comment on their experience completing the study. One participant mentioned “There was a lot of memory recall which wasn’t ideal, I’m glad I made plenty of notes”. In the real-word, it is unlikely someone is reading isolated sentences or making notes as they read a news article. To maintain overall data quality, we included an item for participants to self-report inattentive responding (Meade & Craig, 2012) and while there are few differences between in-person and online data collection, online participants do report more distractions (Clifford & Jerit, 2014). This means we are relatively confident in the overall quality of responses across the two samples, but the methods potentially have low realism. The next step in this line of research is for future studies to capture more real-life behaviour using realistic articles on a wider range of topics such as medicine and finance. Instead of relying solely on self-report data, the design could also use more direct measures such as eye-tracking to monitor how participants interact with the materials. For instance, Lee et al. (2024) compared gaze time on mock Facebook posts about food crises between different types of corrective rebuttals.

Conclusion

Taken together, our study demonstrated a successful direct replication where debunking statistical misinformation reduced mistaken causal inferences. Through the open science practices of Irving et al. (2022) sharing their data and materials, we could computationally reproduce their analyses, replicate their findings with new data, and show the robustness

and generalibility of their conclusions to alternative analysis techniques. Collectively, Irving et al. (2022) and our findings support previous research and theory demonstrating the effectiveness of debunking techniques. Future research can address limitations around external validity to demonstrate whether the effectiveness of correcting statistical misinformation can generalise to more diverse samples and more realistic materials.

Disclosures

Conflicts of Interest

We declare no conflict of interest associated with the publication of this manuscript.

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Open Research Practices

Open data: <https://osf.io/jpxbu/>. Our direct replication data are available under a CC-BY-4.0 licence and the original data from Irving et al. are publicly available on the OSF.

Open materials: <https://osf.io/jpxbu/>

Analysis script: <https://osf.io/jpxbu/>

Pre-registration: This study was not pre-registered.

R Packages

We used R (Version 4.6.0; R Core Team, 2026) and the R-packages *BayesFactor* (Version 0.9.12.4.8; Morey & Rouder, 2023), *bayesplot* (Version 1.15.0; Gabry et al., 2019), *bayestestR* (Version 0.18.1; Makowski et al., 2019), *brms* (Version 2.23.0; Bürkner, 2017),

497 *cowplot* (Version 1.2.0; Wilke, 2020), *dplyr* (Version 1.2.1; Wickham, François, et al., 2023),
498 *effectsize* (Version 1.0.2; Ben-Shachar et al., 2020), *emmeans* (Version 2.0.3; Lenth, 2023),
499 *ggplot2* (Version 4.0.3; Wickham, 2016), *janitor* (Version 2.2.1; Firke, 2023), *knitr* (Version
500 1.51; Xie, 2015), *papaja* (Version 0.1.5; Aust & Barth, 2023), *patchwork* (Version 1.3.2;
501 Pedersen, 2023), *performance* (Version 0.17.0; Lüdtke et al., 2021), *pwr* (Version 1.3.0;
502 Champely, 2020), *qualtRics* (Version 3.2.2; Ginn et al., 2022), *readr* (Version 2.2.0; Wickham,
503 Hester, et al., 2023), *tidybayes* (Version 3.0.7; Kay, 2023), *tidyr* (Version 1.3.2; Wickham,
504 Vaughan, et al., 2023), and *tidyverse* (Version 2.0.0; Wickham et al., 2019) for all our
505 analyses.

References

- Adams, R. C., Challenger, A., Bratton, L., Boivin, J., Bott, L., Powell, G., Williams, A., Chambers, C. D., & Sumner, P. (2019). Claims of causality in health news: A randomised trial. *BMC Medicine*, 17(1), 91. <https://doi.org/10.1186/s12916-019-1324-7>
- Alister, M., Vickers-Jones, R., Sewell, D. K., & Ballard, T. (2021). How Do We Choose Our Giants? Perceptions of Replicability in Psychological Science. *Advances in Methods and Practices in Psychological Science*, 4(2), 25152459211018199. <https://doi.org/10.1177/25152459211018199>
- Aust, F., & Barth, M. (2023). *papaja: Prepare reproducible APA journal articles with R Markdown*. <https://github.com/crsh/papaja>
- Ben-Shachar, M. S., Lüdtke, D., & Makowski, D. (2020). effectsize: Estimation of effect size indices and standardized parameters. *Journal of Open Source Software*, 5(56), 2815. <https://doi.org/10.21105/joss.02815>
- Bürkner, P.-C. (2017). brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software*, 80(1), 1–28. <https://doi.org/10.18637/jss.v080.i01>
- Champely, S. (2020). *Pwr: Basic functions for power analysis*. <https://CRAN.R-project.org/package=pwr>
- Chan, M.-P. S., Jones, C. R., Hall Jamieson, K., & Albarracín, D. (2017). Debunking: A Meta-Analysis of the Psychological Efficacy of Messages Countering Misinformation. *Psychological Science*, 28(11), 1531–1546. <https://doi.org/10.1177/0956797617714579>
- Clarke, B., Alley, L. J., Ghai, S., Flake, J. K., Rohrer, J. M., Simmons, J. P., Schiavone, S. R., & Vazire, S. (2024). Looking our limitations in the eye: A call for more thorough and honest reporting of study limitations. *Social and Personality Psychology Compass*, 18(7), e12979. <https://doi.org/10.1111/spc3.12979>
- Clarke, B., Lee, P. Y., Schiavone, S. R., Rhemtulla, M., & Vazire, S. (2026). The prevalence of direct replication articles in top-ranking psychology journals. *American Psychologist*, 81(2), 238–253. <https://doi.org/10.1037/amp0001385>

- Clifford, S., & Jerit, J. (2014). Is There a Cost to Convenience? An Experimental Comparison of Data Quality in Laboratory and Online Studies. *Journal of Experimental Political Science*, 1(2), 120–131. <https://doi.org/10.1017/xps.2014.5>
- Delaney, C. (2005). The Spirituality Scale: Development and Psychometric Testing of a Holistic Instrument to Assess the Human Spiritual Dimension. *Journal of Holistic Nursing*, 23(2), 145–167. <https://doi.org/10.1177/0898010105276180>
- Ecker, U. K. H., Lewandowsky, S., Cook, J., Schmid, P., Fazio, L. K., Brashier, N., Kendeou, P., Vraga, E. K., & Amazeen, M. A. (2022). The psychological drivers of misinformation belief and its resistance to correction. *Nature Reviews Psychology*, 1(1), 13–29. <https://doi.org/10.1038/s44159-021-00006-y>
- Fasce, A., Rosales-Trabuco, J., Barberia, I., Pinedo-Yzaguirre, E., Espinoza Nuñez, J. M., Ortiz-Treviños, W. M., Pizarro-Moncada, C. D., Reyes-Bossio, M., Carbajal-León, C., & Rodríguez-Ferreiro, J. (2026). Effects of debunking interventions on endorsement of alternative medicine: A randomized controlled experiment in Peru. *Scientific Reports*, 16(1), 4995. <https://doi.org/10.1038/s41598-026-38260-w>
- Field, S. M., Hoekstra, R., Bringmann, L., & Ravenzwaaij, D. van. (2019). When and Why to Replicate: As Easy as 1, 2, 3? *Collabra: Psychology*, 5(1), 1–15. <https://doi.org/10.1525/collabra.218>
- Firke, S. (2023). *Janitor: Simple tools for examining and cleaning dirty data*. <https://CRAN.R-project.org/package=janitor>
- Gabry, J., Simpson, D., Vehtari, A., Betancourt, M., & Gelman, A. (2019). Visualization in bayesian workflow. *Journal of the Royal Statistical Society A*, 182, 389–402. <https://doi.org/10.1111/rssa.12378>
- Ghai, S., Forscher, P. S., & Chuan-Peng, H. (2024). Big-team science does not guarantee generalizability. *Nature Human Behaviour*, 8(6), 1053–1056. <https://doi.org/10.1038/s41562-024-01902-y>
- Ginn, J., O'Brien, J., & Silge, J. (2022). *qualtRics: Download 'qualtrics' survey data*.

<https://CRAN.R-project.org/package=qualtRics>

Hameleers, M. (2023). Disinformation as a context-bound phenomenon: Toward a conceptual clarification integrating actors, intentions and techniques of creation and dissemination. *Communication Theory*, 33(1), 1–10. <https://doi.org/10.1093/ct/qtac021>

Irving, D., Clark, R. W. A., Lewandowsky, S., & Allen, P. J. (2022). Correcting statistical misinformation about scientific findings in the media: Causation versus correlation.

Journal of Experimental Psychology. Applied, 28(2), 1–9.

<https://doi.org/10.1037/xap0000408>

Isager, P. M., Aert, R. C. M. van, Bahník, Š., Brandt, M. J., DeSoto, K. A., Giner-Sorolla, R., Krueger, J. I., Perugini, M., Ropovik, I., Veer, A. E. van 't, Vranka, M., & Lakens, D. (2023). Deciding what to replicate: A decision model for replication study selection under resource and knowledge constraints. *Psychological Methods*, 28(2), 438–451.

<https://doi.org/10.1037/met0000438>

Jang, J., Lee, E.-J., & Shin, S. Y. (2019). What Debunking of Misinformation Does and Doesn't. *Cyberpsychology, Behavior, and Social Networking*, 22(6), 423–427.

<https://doi.org/10.1089/cyber.2018.0608>

Kay, M. (2023). *tidybayes: Tidy data and geoms for Bayesian models*.

<https://doi.org/10.5281/zenodo.1308151>

Klein, R. A., Vianello, M., Hasselman, F., Adams, B. G., Adams, R. B., Alper, S., Aveyard, M., Axt, J. R., Babalola, M. T., Bahník, Š., Batra, R., Berkics, M., Bernstein, M. J., Berry, D. R., Bialobrzeska, O., Binan, E. D., Bocian, K., Brandt, M. J., Busching, R., ... Nosek, B. A. (2018). Many Labs 2: Investigating Variation in Replicability Across

Samples and Settings. *Advances in Methods and Practices in Psychological Science*, 1(4), 443–490. <https://doi.org/10.1177/2515245918810225>

LeBel, E. P., McCarthy, R. J., Earp, B. D., Elson, M., & Vanpaemel, W. (2018). A Unified Framework to Quantify the Credibility of Scientific Findings. *Advances in Methods and Practices in Psychological Science*, 1(3), 389–402.

<https://doi.org/10.1177/2515245918787489>

LeBel, E. P., Vanpaemel, W., Cheung, I., & Campbell, L. (2019). A Brief Guide to Evaluate Replications. *Meta-Psychology*, 3, 1–9. <https://doi.org/10.15626/MP.2018.843>

Lee, Y.-I., Mu, D., Hsu, Y.-C., Wojdyski, B. W., & Binford, M. (2024). Misinformation or hard to tell? An eye-tracking study to investigate the effects of food crisis misinformation on social media engagement. *Public Relations Review*, 50(4), 102483.

<https://doi.org/10.1016/j.pubrev.2024.102483>

Lenth, R. V. (2023). *Emmeans: Estimated marginal means, aka least-squares means*.

<https://CRAN.R-project.org/package=emmeans>

Lewandowsky, S., Ecker, U. K. H., Seifert, C. M., Schwarz, N., & Cook, J. (2012).

Misinformation and Its Correction: Continued Influence and Successful Debiasing.

Psychological Science in the Public Interest, 13(3), 106–131.

<https://doi.org/10.1177/1529100612451018>

Loomba, S., Figueiredo, A. de, Piatek, S. J., Graaf, K. de, & Larson, H. J. (2021).

Measuring the impact of COVID-19 vaccine misinformation on vaccination intent in the UK and USA. *Nature Human Behaviour*, 5(3), 337–348.

<https://doi.org/10.1038/s41562-021-01056-1>

Lopez, A., Choi, A. K., Dellawar, N. C., Cullen, B. C., Avila Contreras, S., Rosenfeld, D. L., & Tomiyama, A. J. (2023). Visual cues and food intake: A preregistered replication of

Wansink et al (2005). *Journal of Experimental Psychology: General*.

<https://doi.org/10.1037/xge0001503.supp>

Lorko, M., Čavojová, V., Šrol, J., Priesol, R., Jalakšová, P., & Tužilová, B. (2026). Timing Matters: The Effects of Prebunking Versus Debunking on Trust in Disinformation.

Personality and Social Psychology Bulletin, 01461672251411571.

<https://doi.org/10.1177/01461672251411571>

Lüdecke, D., Ben-Shachar, M. S., Patil, I., Waggoner, P., & Makowski, D. (2021).

performance: An R package for assessment, comparison and testing of statistical models.

- 614 *Journal of Open Source Software*, 6(60), 3139. <https://doi.org/10.21105/joss.03139>
- 615 Makel, M. C., Plucker, J. A., & Hegarty, B. (2012). Replications in Psychology Research:
616 How Often Do They Really Occur? *Perspectives on Psychological Science*, 7(6), 537–542.
617 <https://doi.org/10.1177/1745691612460688>
- 618 Makowski, D., Ben-Shachar, M. S., & Lüdtke, D. (2019). bayestestR: Describing effects and
619 their uncertainty, existence and significance within the bayesian framework. *Journal of*
620 *Open Source Software*, 4(40), 1541. <https://doi.org/10.21105/joss.01541>
- 621 Meade, A. W., & Craig, S. B. (2012). Identifying careless responses in survey data.
622 *Psychological Methods*, 17(3), 437–455. <https://doi.org/10.1037/a0028085>
- 623 Morey, R. D., & Rouder, J. N. (2023). *BayesFactor: Computation of bayes factors for*
624 *common designs*. <https://CRAN.R-project.org/package=BayesFactor>
- 625 Nadelson, L., Jorcyk, C., Yang, D., Jarratt Smith, M., Matson, S., Cornell, K., & Husting, V.
626 (2014). I Just Don't Trust Them: The Development and Validation of an Assessment
627 Instrument to Measure Trust in Science and Scientists. *School Science and Mathematics*,
628 114(2), 76–86. <https://doi.org/10.1111/ssm.12051>
- 629 National Academies of Sciences, Engineering, and Medicine. (2019). *Reproducibility and*
630 *Replicability in Science*. <https://doi.org/10.17226/25303>.
- 631 Nosek, B. A., & Errington, T. M. (2017). Making sense of replications. *eLife*, 6, e23383.
632 <https://doi.org/10.7554/eLife.23383>
- 633 Park, S., & Nan, X. (2026). Generative AI and misinformation: A scoping review of the role
634 of generative AI in the generation, detection, mitigation, and impact of misinformation.
635 *AI & Society*, 41(2), 1501–1515. <https://doi.org/10.1007/s00146-025-02620-3>
- 636 Pedersen, T. L. (2023). *Patchwork: The composer of plots*.
637 <https://CRAN.R-project.org/package=patchwork>
- 638 R Core Team. (2026). *R: A language and environment for statistical computing*. R
639 Foundation for Statistical Computing. <https://doi.org/10.32614/R.manuals>
- 640 Rouder, J. N., Speckman, P. L., Sun, D., Morey, R. D., & Iverson, G. (2009). Bayesian t

tests for accepting and rejecting the null hypothesis. *Psychonomic Bulletin & Review*,
16(2), 225–237. <https://doi.org/10.3758/PBR.16.2.225>

Schmid, P. (2026). Debunking health misinformation with empathy. *Current Opinion in Psychology*, 67, 102213. <https://doi.org/10.1016/j.copsyc.2025.102213>

Sumner, P., Vivian-Griffiths, S., Boivin, J., Williams, A., Venetis, C. A., Davies, A., Ogden, J., Whelan, L., Hughes, B., Dalton, B., Boy, F., & Chambers, C. D. (2014). The association between exaggeration in health related science news and academic press releases: Retrospective observational study. *BMJ*, 349, g7015. <https://doi.org/10.1136/bmj.g7015>

The Turing Way Community. (2025). *The Turing Way: A handbook for reproducible, ethical and collaborative research*. Zenodo. <https://doi.org/10.5281/zenodo.15213042>

Van de Schoot, R., Depaoli, S., King, R., Kramer, B., Märtens, K., Tadesse, M. G., Vannucci, M., Gelman, A., Veen, D., Willemsen, J., & Yau, C. (2021). Bayesian statistics and modelling. *Nature Reviews Methods Primers*, 1(1), 1–26. <https://doi.org/10.1038/s43586-020-00001-2>

Wickham, H. (2016). *ggplot2: Elegant graphics for data analysis*. Springer-Verlag New York. <https://ggplot2.tidyverse.org>

Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemund, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., . . . Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43), 1686. <https://doi.org/10.21105/joss.01686>

Wickham, H., François, R., Henry, L., Müller, K., & Vaughan, D. (2023). *Dplyr: A grammar of data manipulation*. <https://CRAN.R-project.org/package=dplyr>

Wickham, H., Hester, J., & Bryan, J. (2023). *Readr: Read rectangular text data*. <https://CRAN.R-project.org/package=readr>

Wickham, H., Vaughan, D., & Girlich, M. (2023). *Tidyr: Tidy messy data*.

668 <https://CRAN.R-project.org/package=tidyr>

669 Wilke, C. O. (2020). *Cowplot: Streamlined plot theme and plot annotations for 'ggplot2'*.

670 <https://CRAN.R-project.org/package=cowplot>

671 Xie, Y. (2015). *Dynamic documents with R and knitr* (2nd ed.). Chapman; Hall/CRC.

672 <https://yihui.org/knitr/>

673 Xu, G., Qian, M., & Meng, L. (2025). Misinformation dissemination on social media: Key
674 research themes and evolutionary paths between 2013 and 2023. *Humanities and Social
675 Sciences Communications*, 12(1), 1775. <https://doi.org/10.1057/s41599-025-06067-1>

676 Zwaan, R. A., Etz, A., Lucas, R. E., & Donnellan, M. B. (2018). Making replication
677 mainstream. *Behavioral and Brain Sciences*, 41, e120.

678 <https://doi.org/10.1017/S0140525X17001972>